

Introduction to Object-Oriented Programming

CS 258 | Number of Credits: 3 | Fall 2025

Instructor: Dr. Srishti Srivastava

Class meeting: M,W,F 11:00 – 11:50 AM

Office Hour: by appointment only

Classroom: BE 1034

Email: fsrishti@usi.edu

Phone: 812-228-5032

Office: BEC 2085

Zoom Meeting ID (for any online sessions): 993-8308-0901

Course Description

This course offers an introduction to programming, emphasizing object-oriented programming (OOP) techniques. Students will learn the foundational elements of the chosen language, including syntax, data types, methods, conditionals, iterations, functions, and procedures, all essential for crafting object-oriented programs. The curriculum will begin with console-based examples to solidify core concepts, gradually transitioning to graphical user interface (GUI) examples as the semester progresses. Practical laboratory exercises will be assigned throughout the course, designed to simulate common science and business problems for which the programming language is particularly well-suited, providing hands-on experience in real-world applications. This course, in particular, includes instruction in the following knowledge units identified in the ACM Computer Science CS2023 Knowledge Units:

1. Object-oriented design:
 - a. Decomposition into objects carrying state and having behavior.
 - b. Class-hierarchy design for modeling.
2. Definition of classes: fields, methods, and constructors
3. Dynamic vs static properties.
4. Collection classes, iterators, and other common library components

Prerequisites: None

Course Learning Objectives

Upon successful completion of this course, students will be able to:

1. Apply Introductory Level OOP Concepts:
 - Demonstrate proficiency in fundamental object-oriented programming, including control structures, data types, and methods.
 - Construct a class through design, implementation, and testing to meet behavioral requirements
2. Understand Object-Oriented Programming Principles:
 - Explain and apply key principles of object-oriented programming such as collection classes and iterators effectively to solve a problem.
 - Characterize classes and objects to create modular and reusable code.
3. Analyze Design Patterns:
 - Identify and implement common design patterns to solve programming challenges.
 - Explore the importance of design patterns in creating scalable and maintainable software.

4. Understand Parallel Computing Fundamentals

- Explain Sequential vs. parallel computing, data parallelism, speedup, and parallel overhead.
- Apply existing parallel methods to optimize sequential code.

5. Apply Proficient Debugging Skills:

- Employ effective debugging techniques to identify and resolve programming errors.
- Utilize debugging tools and practices to troubleshoot and optimize code.

6. Create Multithreaded Applications:

- Design and implement multithreaded applications.
- Outline the challenges and best practices associated with concurrent programming.

7. Collaborate on Group Projects:

- Work collaboratively in a team environment on a group project.
- Communicate effectively with team members, share code, and integrate individual contributions into a cohesive application.

8. Practice Code Reviews and Documentation:

- Participate in code reviews to provide and receive constructive feedback.
- Employ code documentation for enhancing code maintainability.

9. Develop a Final Project:

- Plan, design, and implement a comprehensive final project using OOP principles.
- Present the final project to the class, demonstrating the application of learned concepts in a real-world scenario.

10. Cultivate Problem-Solving Skills:

- Generate solutions to complex programming challenges using a systematic and logical approach.
- Develop the ability to break down problems into manageable tasks and implement effective solutions.

Note: the above measurable action verbs are based on Bloom's Taxonomy.

Program Learning Outcomes

This course supports the expected characteristics, capabilities, and skills for computer science graduates in the USI Computer Science program of study in the following ways:

1) Mastery of CS technical foundations. Students will begin to build knowledge and understanding of computer science areas including algorithms and design, fundamental programming concepts, fundamental data structures, software development methods, and ethics.

2) Recognition of common CS themes and principles. Students will begin to be able to recognize and identify recurring themes that present themselves in many diverse computer science domains, recognize and identify recurring principles that present themselves in many diverse computer science domains, and apply recurring themes and principles where appropriate.

3) Recognition of interplay between theory and practice. Students will begin to be able to explain and illustrate how theory and practice influence each other.

4) Effective problem solving and critical thinking skills. Students will begin to be able to identify and use relevant concepts, identify and use relevant information, select and use appropriate actions or operations, interpret clearly and logically from prior activities,

develop qualitative and quantitative assessment and develop solutions to problems not only in single levels of details, but also multiple levels of detail and abstraction.

5) Commitment to life-long learning, and professional and ethical responsibility. Students will be introduced to the concept that they must develop a plan that establishes how they will maintain skills to remain relevant in the CS field, be able to identify cross disciplinary opportunities between CS and other fields, and to recognize, analyze, and address the intellectual, professional, economic, social, legal, ethical, and cultural issues associated with computing.

6) Awareness of the broad applicability of computing. Students will be introduced to a wide range of opportunities available in computing.

7) Awareness of domain-specific knowledge. Students will be introduced the interactions between CS and many different domains and will begin to be able to communicate with, and learn from, experts from different domains.

Communication Guidelines

- **Netiquette**

All class communications and interactions with other students and the professor should follow common social standards for respect and courtesy. Learn about the [USI Netiquette Guidelines for Online Students](#) so that you can get the most out of your class.

- **Communication Expectation**

For questions related to the course material, assignments, exams, etc., students may contact the instructor in person in class, or during the listed office hours, or via email/blackboard discussion forums. The students should expect to receive a response from the instructor to their email within 48 hours. Students should expect to receive feedback on exams and assignments within two weeks after due date.

Blackboard Course Site Overview

The blackboard course site will be organized into multiple course tabs provided on the left panel of the web page. The main course tabs include the following information.

1. *Content*: this tab contains course materials such as video lectures, slides, informational links, assignments, quizzes, tests, and others.
 - a. *Start here*: this tab provides information about the course syllabus, instructor, further instructions to navigate the course, etc.
2. *Calendar*: this tab contains information on listed due dates for assignments, quizzes, tests, holidays, etc.
3. *Announcements*: Throughout the semester, this tab provides course related announcements such as assignment due dates, scheduled exams, syllabus changes, etc.

Course Requirements

- **Textbooks and/or Course Materials**

Tony Gaddis. 2022. *Starting Out with Java: Control Structures through Objects* (8th ed.). Pearson (Revel).

- **Technical Requirements**

- **A Reliable Computer:** a dedicated computer with an updated operating system, such as Windows 7 or later, or Mac OSX.
- **High-Speed Internet Connection:** (e.g. DSL or Cable). All USI online students are required to have a stable high-speed Internet connection. A wired Internet connection is recommended for online meetings, exams, and assignment submission.
- [Office 365](#) (USI login needed) & [Adobe Acrobat Reader](#) installed.
- **Supported Web Browser:** [Firefox](#), [Google Chrome](#), or [Safari](#).
- Microphone & Webcam
- VS Code or NetBeans IDE (or a similar IDE for Java)

* More information on the system requirements for taking online courses can be found at [USI Online Learning System Requirements](#)

Minimum Technical Skills

Navigating the Blackboard; Commenting and sharing VoiceThread presentation; Basic software installation.

Minimum Digital Information Literacy Skills

1. Using online libraries and databases to locate and gather appropriate information
2. Using computer networks to locate and store files or data
3. Using online search tools for specific academic purposes, including the ability to use search criteria, keywords, and filters
4. Properly citing information sources
5. Preparing a presentation of research findings
6. Uploading and downloading appropriate course related files to/from blackboard.

Assessments

Assignments- There will be assignments for every course module in this course that are worth 100 points each. Some of the assignments you may choose to work in a group. Pay close attention to the assignment details on the Blackboard site within the corresponding modules. Rubrics for these assignments are also located on the Blackboard site.

Quizzes- several quizzes in this course that assess your knowledge with the corresponding module's information. Most of these questions are multiple choice, T/F, and/or matching, with a couple short answer/essay questions. You will be given 15 minutes to complete these short

assessments. Quizzes are worth 15, 20, 20, and 25 points respectively. See more information on the Blackboard site.

Exams- There are three exams in this course. Exam 1, Exam 2, and the Comprehensive Final Exam. The final exam may include a course (team based) project and a final presentation. Please read the exam instructions on the Blackboard site. Each Exam is worth 100 points.

Lecture Responses/Discussions: Within most of the modules, there will be opportunities to interact with the content, your peers, and your instructor. Within specific lectures, there will be questions to assess your knowledge and/or understanding of the concepts. These are self-check opportunities and will not be counted towards a grade, but you are highly encouraged to participate and/or to respond via text, audio, and/or video to your peers to assess your own understanding of those topics.

Grading Methods

Course Deliverables	Point Value/Range per deliverable	Total Percentage
Assignments	100	50%
Quizzes	10	20%
Final Exam/Project	100	30%
TOTAL		100 %

Note: this is a tentative points distribution across course deliverables. This may be subjected to minor changes during the semester based on course progress. Students are expected to follow blackboard and email announcements regarding the course.

	A	B	C	D	F
Percentage	100-90%	89-80%	79-70%	69-60%	Below 59%

Course & University Policies

Participation

To obtain full credit for participation, students are expected to complete all required readings, and review all course material such as, lecture slides, provided text and online links. Students are also expected to provide all required VoiceThread responses, attend scheduled face-to-face or Zoom meetings, complete all assignments, quizzes, and exams. You are expected to spend 6 hours a week for lectures, course readings, assignments, and exams. You are also expected to log in to the course at least 3 times in a week.

Late Assignment Policy

Assignments are due at the end of the day due. Any assignment submitted after the due date will be considered late and will be accepted only with a penalty of one letter grade for each day (or partial day) late.

Academic Integrity

Students are expected to maintain complete honesty and integrity in the academic experiences both in and out of the classroom. Any student found guilty of dishonesty including, but not limited to, plagiarism and/or cheating on an examination will be subject to disciplinary action. For more information, please visit [Academic Integrity Policies and Procedures](#)

SafeAssign

Plagiarism and cheating constitute academic misconduct and can result in both a grade penalty imposed by the instructor and disciplinary action including suspension or expulsion. As part of their responsibility to uphold the Student Code of Conduct, instructors reserve the right to have papers submitted through SafeAssign to check for plagiarism against a database of papers submitted previously at USI, a national database of papers, and the Internet. Self-plagiarism can also occur when you plagiarize yourself if you submit for credit a piece of work that is the same or substantially similar to work for which you have already gained or intend to gain university credit, at this or any other university. To avoid self-plagiarism, you must have prior permission from the relevant instructor(s), and give full attribution to the source (i.e. yourself). Safe Assign can be used to check against any earlier submitted student assignments or presentations.

Civility Statement

The University of Southern Indiana embraces and celebrates the many differences that exist among the members of a dynamic, intellectual, and inclusive community, and strives to maintain an environment that respects differences and provides a sense of belonging and inclusion for everyone. Any form of discrimination or harassment, including sexual harassment and sexual misconduct, will not be tolerated. All members of the University community are expected to conduct themselves in a manner that does not infringe upon the rights of others, and should be aware that the University is prepared to take prompt action to prevent and correct such inappropriate behavior and to remedy its effects. For more information, please visit <https://handbook.usi.edu/sexual-harassment-policy>

Disability Statement

If you have a disability for which you may require academic accommodations for this class, please contact Disability Resources at 812-464-1961 or email Disability Communications at usi1disres@usi.edu as soon as possible. Students are approved for accommodations by Disability Resources should request their accommodation letter be emailed to them to forward to their online instructors. Due to the nature of online courses, some accommodations approved for on campus courses may not apply. Please discuss this with Disability Resources to clarify as needed. Students who receive an accommodation letter from Disability Resources are encouraged to discuss the provisions of those accommodations with their professors before or during the first week of the semester. If you will be in an internship, field, clinical, student teaching, or other off-campus setting this semester please note that approved academic accommodations needed for access while in this setting. For more information, please visit the Disability Resources website at <http://www.usi.edu/disabilities>.

Class Withdrawal & Incomplete Policy

It is the student's responsibility to officially drop/withdraw from any courses before the deadline. The university does not withdraw students from any classes. Please refer to the USI Academic Calendar (<https://www.usi.edu/registrar/academic-calendar>) for specific dates. For more information, please visit [Registrar's Office Schedule Changes](#). Under special circumstances, students may petition for an incomplete grade. However, it is up to the course instructor to decide if an incomplete will be granted.

Students receiving an incomplete grade will need to complete all course requirements by the agreed deadline to avoid an “F” grade.

Title IX, Sexual Assault and Gender Violence Policy

USI does not tolerate acts of sexual misconduct, including sexual harassment and all forms of sexual violence. If you have experienced sexual misconduct, or know someone who has, the University can help. It is important to know that federal regulations and University policy require faculty to promptly report incidences of potential sexual misconduct known to them to the Title IX Coordinator to ensure that appropriate measures are taken and resources are made available. The University will work with you to protect your privacy by sharing information with only those who need to know to ensure we can respond and assist. If you are seeking help and would like to speak to someone confidentially, you can make an appointment with a counselor in the University Counseling Center. Find more information about sexual violence, including campus and community resources at <https://www.usi.edu/institutional-equity/reporting-options>

Technical Support

Help Desk Support

Since your professors are not trained as technology experts, they will not be able to assist you with your technical needs. If you have any issues with Blackboard, Zoom, Panopto, VoiceThread, or any other technology, please contact the [Help Desk](#) at (812) 465-1080 or send an email to it@usi.edu. To learn more about the USI helpdesk, please visit [USI IT Help Desk](#)

Additional Training Documentation

- [Blackboard](#)
- [Zoom](#)
- [Panopto](#)

Privacy Policies & Accessibility Statements

For detailed information about the privacy policies and accessibility statements of the course technologies, please visit [USI Online Learning Support Services](#)

Academic Support Services

- **Academic Skills**

<http://www.usi.edu/university-division/academic-skills/tutoring/>

This office offers tutoring and learning assistance to help you develop skills to succeed in your academic life. To contact them, please call 812/464-1743 or go to the website above to find the latest tutoring schedule. In addition to tutoring in math and other content areas in Academic Skills, please note that the Writers’ Room (ED 1102) provides writing assistance in person and online. To use this service, please contact their office by calling 812/461-5359 or visit their website at <https://www.usi.edu/university-division/academic-success-center/tutoring>

- **How to Cite Sources (IEEE, ACM, APA, MLA, Chicago, Turabian, etc.)**

<http://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>

<https://www.acm.org/publications/authors/reference-formatting>

<http://usi.libguides.com/citingsources>

- **Online Learning Library Services**
<https://usi.libguides.com/onlinelearning>

- **Counseling Center**

The Counseling Center is open every Monday through Friday from 8:00 am to 4:30 pm. You'll find the Counseling Center in the Orr Center, Room 1051; their phone number is 812/464-1867

<https://www.usi.edu/counseling-and-psychological-services>

- **Disability, and ADA Support**

<https://www.usi.edu/disability-resources>

- **Student Grievances**

<https://handbook.usi.edu/student-academic-grievance-procedure>

- **Additional Student Resources**

The university provides a comprehensive range of support services and activities for students.

Please refer to <http://www.usi.edu/students/> for detailed information regarding academic advising, registration, financial aid, student affairs, counseling, career services, etc.

- **University Health Center**-The University Health Center is a full-service clinic offering medical services and health-related information to students, faculty, and staff. It located in the lower level of the Health Professions Center and is open Monday through Friday, 8 am to 4:30 pm. Click on this link to learn more about the [Health Services](#) offered at USI, or to make an appointment, please call 812/465-1250..
- **Financial Aid Office**- Information about current or future financial aid (including the implications of withdrawal from courses) can be obtained from the University Financial Aid Office. The office open is every Monday through Friday from 8:00 am to 4:30 pm; phone 812/464-1767, or email them at finaid@usi.edu
- **Student Basic Needs**
Students who have difficulty affording food on a regular basis or lack a safe place to live and believe this may affect their class performance are encouraged to contact the [Dean of Students Office](#). A list of resources can also be found at <https://www.usi.edu/financial-success/financial-wellness>

Online Materials and Recordings Statements

Course Materials

My lectures and course materials, including presentations, tests, exams, outlines, and similar materials, are protected by copyright. I am the exclusive owner of copyright in those materials I create. I encourage you to take notes and make copies of course materials for your own educational use. However, you may not, nor may you knowingly allow others to reproduce or distribute lecture notes and course materials publicly without my express written consent. This includes providing materials to commercial course material suppliers such as Course Hero and other similar services. Students who publicly distribute or display or help others publicly distribute or display copies or modified copies of an instructor's course materials may be in violation of The Student Rights and Responsibilities: A Code of Student Behavior. Similarly, you own copyright in your original papers and exam essays. If I am interested in posting your answers or papers on the course web site, I will request your written permission.

Recorded Class Sessions

Some of the sessions in this course may be recorded or live-streamed. Such recordings/ streaming will only be made available to students registered for this class. Students may not share these sessions with others not in the class or upload them to any other online environment. Doing so would be a breach of the Code of Student Conduct, and, in some cases, a violation of the Federal Education Rights and Privacy Act (FERPA).